The IMO ship recycling regulations
A perspective

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Vocabulary

- **SHIP DISMANTLING**
  used by Basel Convention; by Ministries of Environment; and by European Commission

- **SHIP BREAKING**
  used by ILO; and by Environmental NGOs (Greenpeace et al)

- **DEMOLITION** or **DEMO**
  used by brokers

- **DISPOSALS**
  often used in shipping statistics

- **SCRAPPING**
  used by shipowners; and by ILO/IMO/BC JWG

- **SHIP RECYCLING**
  used by IMO and by Ministries of Transport/Shipping

Is it political correctness that makes us say “recycling”?



Is it “recycling”?



Is it “recycling”?



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Is it “recycling”?



… and the proverbial sink



What is the International Maritime Organization

IMO is a specialized agency of the United Nations with 169 Member States and three Associate Members. The meetings of IMO are also attended by observer accredited NGOs and IGOs.

The Convention establishing the International Maritime Organization (IMO) was adopted in Geneva in 1948 and IMO first met in 1959. IMO's main task has been to develop and maintain a comprehensive regulatory framework for shipping and its remit today includes safety, environmental concerns, legal matters, technical co-operation, maritime security and the efficiency of shipping.

IMO is based in the United Kingdom with a Secretariat of around 300 international staff.

How does IMO operate?

The Organization consists of an Assembly, a Council and four main Committees: the Maritime Safety Committee; the Marine Environment Protection Committee; the Legal Committee; and the Technical Co-operation Committee. There is also a Facilitation Committee and a number of Sub-Committees support the work of the main technical committees.

What does IMO produce?

A comprehensive body of international conventions, supported by hundreds of recommendations and guidelines governing every facet of shipping.

There are, firstly, measures aimed at the prevention of accidents, including standards for ship design, construction, equipment, operation and manning. Key treaties include SOLAS, the MARPOL convention for the prevention of pollution by ships and the STCW convention on standards of training for seafarers.

What does IMO produce?

Then, there are measures which recognize that accidents do happen, including rules concerning distress and safety communications, the International Convention on Search and Rescue and the International Convention on Oil Pollution Preparedness, Response and Co-operation.

Thirdly, there are conventions which establish compensation and liability regimes - including the International Convention on Civil Liability for Oil Pollution Damage, the convention establishing the International Fund for Compensation for Oil Pollution Damage and the Athens Convention covering liability and compensation for passengers at sea.

What does IMO produce?

IMO has also an extensive technical co-operation programme, which identifies needs among Members and matches them to assistance, such as training.

Two advanced level maritime educational institutes have been established under the auspices of IMO: WMU in Malmö, and IMLI in Malta.

In addition IMO funds a number of training programs and other capacity building activities (in 2008 spent US$ 10.3M).

How does IMO develop a convention?

The development of a new convention is decided by the Assembly who then requests a Committee to develop the text, occasionally with help from one or more of its Sub-Committees.

The development of a convention can take a long time (an extreme example is the Wreck Removal Convention which took 30 years) and may involve meetings of: Working Groups, Intersessional Working Groups, Correspondence Groups, and Drafting Groups.

How does IMO develop a convention?

The adoption of a convention is by a diplomatic Conference.

The provisions for amendments to a convention are specified in its text.

Once a convention is adopted it cannot be amended until it enters into force.

The entry into force conditions are specified in the text of the convention. Some conventions have taken a long time to enter into force, and, for example, the HNS (Hazardous And Noxious Substances) Convention has still not entered into force 25 years after it was first adopted.

Views of IMO



Views of IMO



Views of IMO



Background to the development of the Hong Kong Convention

- MEPC 42 (November 1998) The issue of ship recycling is first brought to the attention of the IMO Marine Environment Protection Committee
- MEPC 47 (March 2002) agrees that IMO should develop recommendatory guidelines to be adopted by an Assembly resolution
- MEPC 49 (July 2003) finalizes the IMO Guidelines on Ship Recycling
- Assembly 23 adopts on 5 December 2003 the IMO Guidelines on Ship Recycling by resolution A.962(23)
- Assembly 24 adopts on 1 December 2005 resolution A.981(24) instructing MEPC to develop a “new legally binding instrument on ship recycling”

Background to the development of the new international Convention

Resolution A.981(24) states that the new instrument will regulate:

1 the design, construction, operation and preparation of ships so as to facilitate safe and environmentally sound recycling, without compromising their safety and operational efficiency;

2 the operation of ship recycling facilities in a safe and environmentally sound manner; and

3 the establishment of an appropriate enforcement mechanism for ship recycling (certification / reporting requirements).

Background to the development of the new international Convention

- MEPC 54 (20-24 March 2006) convened a working group on ship recycling which developed the draft text which had been submitted by Norway

- Thereafter numerous meetings were arranged for the further development of the draft text of the convention:

Background to the development of the new international Convention

- CG reports to MEPC 55
- MEPC 55 & WG (October 2006)
- 2nd ISWG (May 2007)
- CG reports to MEPC 56
- MEPC 56 & WG (July 2007)
- 3rd ISWG (January 2008)
- CG reports to MEPC 57
- MEPC 57 & WG (March 2008)
- 4th ISWG (September 2008)
- CG reports to MEPC 58
- MEPC 58 & DG (October 2008)
- Diplomatic Conference (May 2009)

3 years and 2 months from the first submission of draft text up to the adoption of the Convention is probably a record.

Interagency co-operation

Ships

in most countries are regulated by ministries of Transport, or Shipping; international regulations for ships are agreed at IMO

Ship recycling facilities

are regulated by ministries of Labour, or Environment, or even Steel; international requirements for ship recycling facilities have been set in the past by ILO, by BC and more recently by IMO

Interagency co-operation

- The International Labour Organization (ILO) is the specialized agency mandated to set standards on occupational safety and health (OSH) for all workers

- ILO has adopted various Conventions and Recommendations, including in March 2004 the ***ILO Guidelines on Safety and Health in Shipbreaking: Guidelines for Asian countries and Turkey***

Interagency co-operation

- ***The Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal*** (BC) was adopted in 1989 and entered into force in 1992. It regulates the transboundary movement of hazardous and other wastes and has 172 Parties.

- In 1994 Parties to the BC adopted the BAN amendment (banning export of hazardous waste from OECD to non-OECD countries). Not in force.

- The BC developed and approved on 13 December 2002 the: ***Technical Guidelines for the Environmentally Sound Management of the Full and Partial Dismantling of Ships***

- The Conference of the Parties to the BC decided in October 2004 (decision VII/26) that "*a ship may become waste as defined in article 2 of the BC and at the same time it may be defined as a ship under other international rules*".

- European Union implemented the BC and the BAN amendment into its Waste Shipment Regulation (***Regulation*** (EC) No 1013/2006 of the European Parliament and of the Council of 14 June 2006) - applicable since July 2007.

Interagency co-operation

- The three Organizations co-operate since 2005 through the Joint ILO/IMO/BC Working Group on Ship Scrapping;
- The aim of the JWG is to avoid duplication of work and overlapping of roles, responsibilities and competencies, and also to identify any further needs. The JWG has not been a forum for a joint development of the Convention on ship recycling (BC and ILO have contributed directly at IMO to the development of the Hong Kong Convention);
- The Joint Working Group has undertaken a comparison of the technical guidelines of the three organizations; has encouraged collaboration in technical cooperation activities; and has identified activities that could promote implementation of the guidelines.

Adoption of the Convention

- The 2009 International Conference on the Safe and Environmentally Sound Recycling of Ships took place at the Hong Kong Convention and Exhibition Centre, from 11 to 15 May 2009.

- The Conference adopted the “Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships, 2009” and also six resolutions.

The diplomatic Conference



The diplomatic Conference



The diplomatic Conference



The diplomatic Conference



The diplomatic Conference



The diplomatic Conference



Structure of the Hong Kong Convention

The Convention includes:

- 21 Articles (explicit amendments' procedure)
- 25 regulations (tacit amendments), divided in four sections:
  - **General**
  - **Requirements for ships**
  - **Requirements for ship recycling facilities**
  - **Reporting requirements**
- 7 Appendices (tacit amendments)

Also 6 voluntary guidelines are currently being developed providing clarifications, interpretations, and uniform procedures for technical issues arising from the provisions of the Convention.

Application of the Hong Kong Convention

The Convention shall **apply** to:

- ships flying the flag of a Party; and
- recycling facilities operating under the jurisdiction of a Party.

The Convention shall **not apply** to:

- Warships; government owned non- commercial ships; exclusively domestically operated ships; and ships of less than 500GT

"No more-favourable treatment" for ships flying the flag of non-Parties

The adopted threshold of 500 gross tons





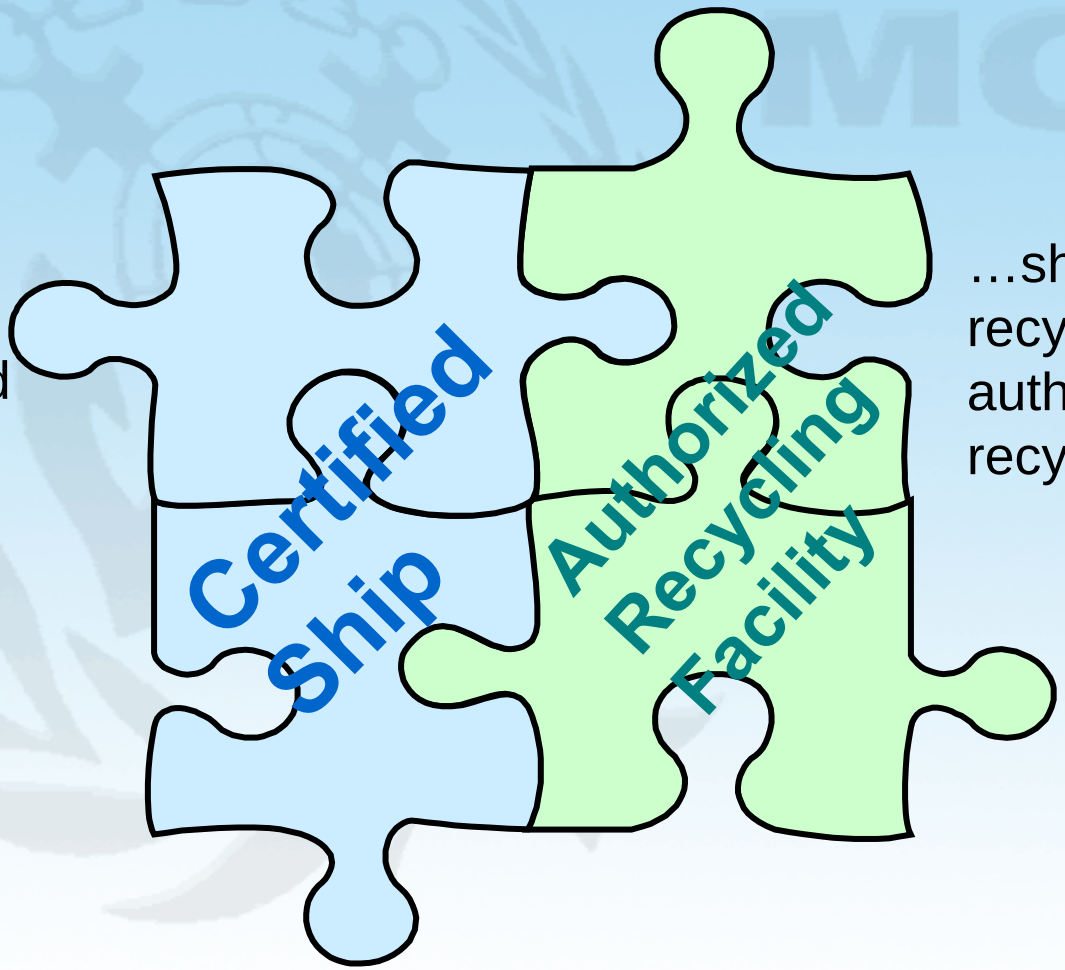




The Hong Kong Convention in a nutshell

Certified ship seeks authorized facility

A ship designed, constructed, operated and prepared for recycling in accordance with the convention...



…should find a recycling facility authorized to recycle that ship

If the Ship Recycling Facility is authorized to handle the hazardous materials identified in the Inventory prior removal shall not be required unless the Ship Recycling Facility decides otherwise when preparing the Ship Recycling Plan.

The Hong Kong Convention in a nutshell

Chain of documentation - "From cradle to grave"

International Certificate on Inventory of Hazardous Materials*

- Inventory of Hazardous Materials Part I (Ship's structure and equipment)
- Ensure that prohibited materials are not used

International Ready for Recycling Certificate

- Ship Recycling Plan
- Inventory Part I, Part II (operational-ly generated waste) & Part III (stores)

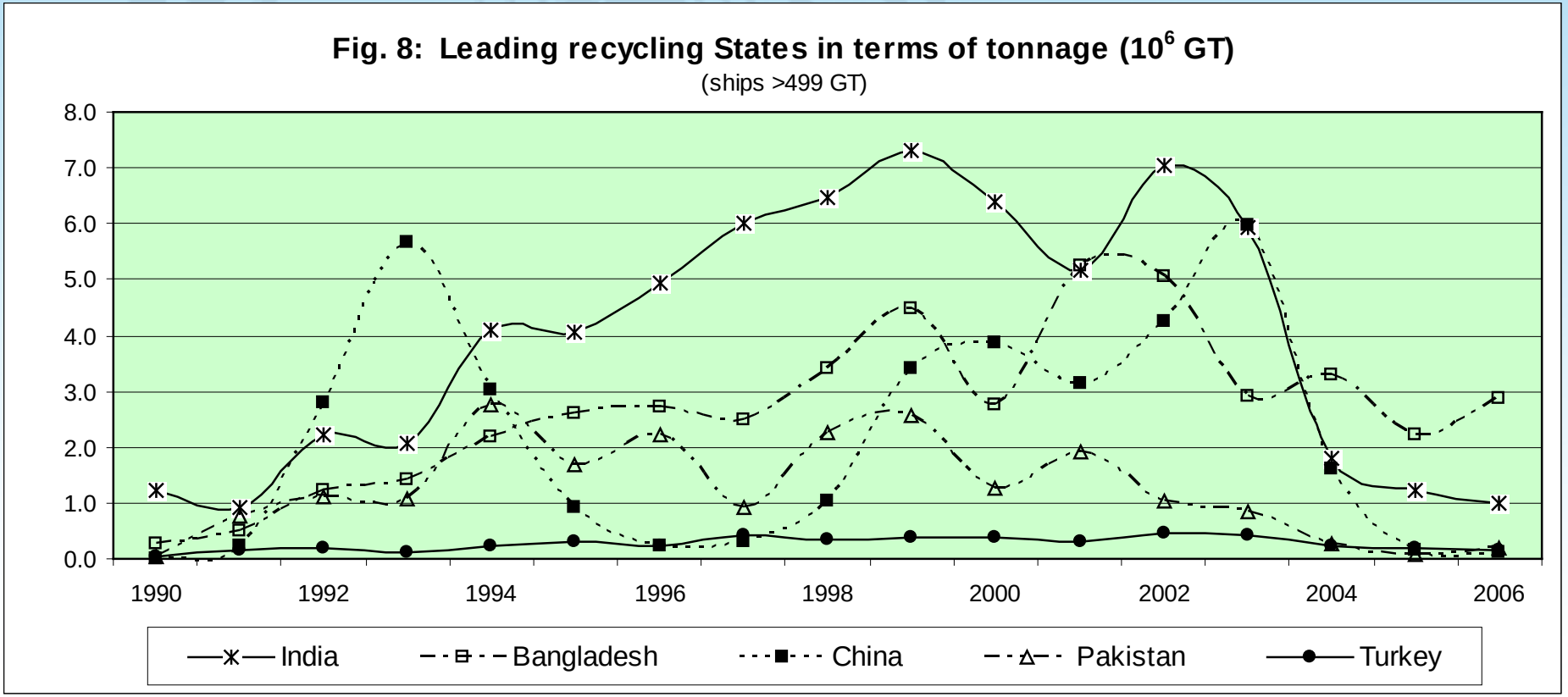
Document of Authorization to conduct Ship Recycling (DASR)

- identifies the capabilities of the facility
- requires Ship Recycling Facility Management Plan

Statement of Completion

*** different requirements for new and existing ships**

The key recycling States



China 12/2007



China 12/2007



China 12/2007



China 12/2007



China 12/2007



China 02/2009



China 02/2009



India 01-2008



India 01-2008



India 01-2008



India 01-2008



India 01-2008



India 01-2008



India 01-2008



India 02-2009



India 02-2009



India 02-2009



India 02-2009



India 02-2009



India 02-2009



Bangladesh (Chittagong) 01-2008



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and last but not least…Bangladesh (Chittagong) 01-2009



The characteristics of ship recycling

in developing countries recycled materials have value;

in developed countries recycled materials do not have much value;

in developed countries the cost of dismantling a ship is high (even when the process is mechanized); and

in developing countries the cost of dismantling a ship is relatively low.

consequently, the natural home of ship recycling is in developing countries.

The characteristics of ship recycling

The implementation of measures for health & safety and for the prevention of pollution is not considered essential in developing countries.

Whereas this applies to all aspects of life in developing countries, the developed world has been concerned about the recycling workers' safety and about the environmental impact of ship recycling.

The Hong Kong Convention certainly addresses these concerns and can introduce a global, practical and uniform standard to the industry.

There are still issues I consider important but which I did not manage to discuss, e.g.:

- differences in the economic drivers in the key recycling countries;
- interim voluntary measures prior to entry into force of the Hong Kong Convention;
- the provisions and the expected date for the entry into force of the Hong Kong Convention;
- the role of the environmental NGOs; and
- a vision of the future.

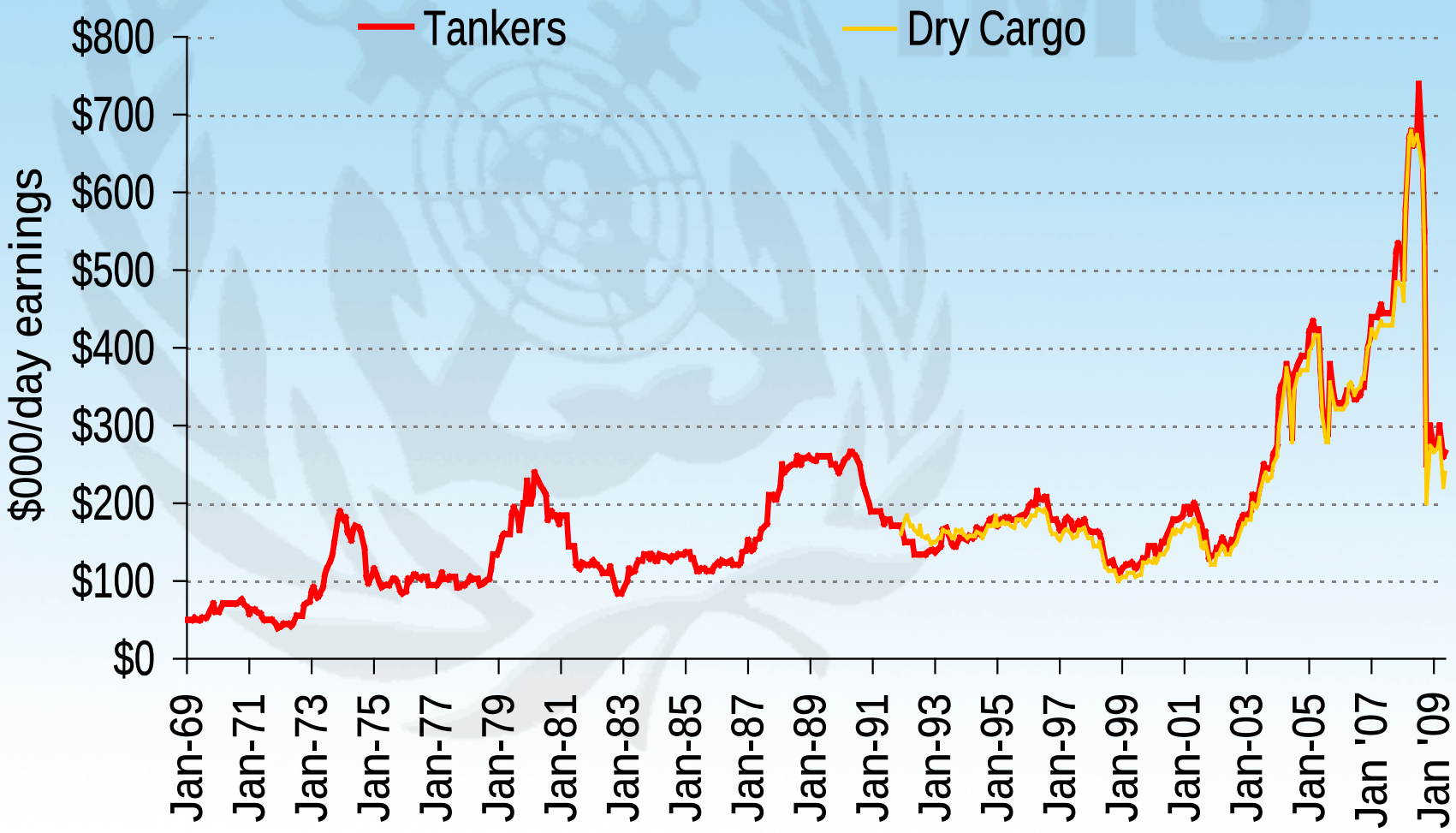
Thank you for your attention






IMO

Demolition Prices



Source: Clarkson Research

A simple equation

Recycler's costs
= price of ship + cost of labour + taxes + costs of compliance with safety and environmental regulations

Recycler's income
= sale of steel scrap + non-ferrous metals + equipment + machinery + furnishings + timber (etc)
+/? subsidies

Note: the cost of labour is not the only parameter that determines a recycler's ability to compete

Table 2 Approximate estimates of lightship content of different ship types

Ship Type	Size (DWT tonnes)	Gross Tonnage	Lightship (long tons)
Tanker VLCC (older)	270,000	146,000	35,000
Tanker VLCC (newer)	300,000	159,000	35,000
Tanker Suezmax	150,000	80,000	22,000
Tanker Aframax	80-120,000	45-67,000	15-18,000
Tanker Panamax	70,000	40,000	10-13,000
Tanker Handysize	35,000	22,000	7,000
Capesize bulkcarrier	150-170,000	78-86,000	20-21,000
Panamax bulkcarrier	70,000	40,000	10-12,000
Handysize bulkcarrier	35,000	22,000	7,000

Table 3 Recycling statistics (ships > 499 GT)

Year of recycling	number of ships (n)	average age (μ)	standard deviation (σ)	Gross Tonnage (GT)	average GT
2006	386	32.6	7.9	4,311,039	64,946
2005	361	32.7	8.9	4,036,787	71,002
2004	615	31.7	8.7	7,478,622	76,885
2003	874	29.7	7.3	16,532,724	85,746
2002	740	28.2	6.2	18,079,566	88,500
2001	772	27.7	5.7	15,903,761	84,537
2000	706	27.3	5.9	14,087,466	87,265
1999	829	26.2	5.4	19,534,461	86,918
1998	801	26.3	5.1	14,343,031	76,259
1997	735	26.5	5.5	10,674,688	79,006
1996	672	26.7	7.1	11,206,752	84,742
1995	649	26.6	6.9	10,226,503	95,645
1994	694	26.6	8.1	12,807,811	98,692
1993	549	26.6	7.8	10,721,544	92,328
1992	450	26.2	7.7	7,566,541	84,435
1991	325	25.7	7.4	3,301,650	73,195
1990	231	25.7	7.8	1,983,758	67,863
TOTALS	**10,389**	**27.7**	**6.9**	**182,796,704**	**17,595**

Table 6 Recycling statistics of major ship types (ships > 499 GT)

Year of recycling	Bulkers		Tankers		Gas ships		Cargo ships		Passenger ships	
	n	μ	n	μ	n	μ	n	μ	n	μ
2006	48	32.1	105	30.4	5	29.8	118	33.5	22	37.5
2005	20	33.3	131	31.2	8	31.3	47	33.5	42	35.8
2004	25	30.8	187	29.6	22	30.0	167	32.7	61	37.9
2003	102	27.4	275	28.6	20	29.1	307	29.9	41	37.2
2002	151	27.0	200	28.5	20	27.9	264	27.5	23	35.0
2001	194	26.6	136	27.8	9	30.3	328	27.2	22	35.0
2000	128	26.8	159	26.9	10	31.1	304	27.2	17	32.5
1999	222	24.8	129	25.4	8	31.9	356	26.3	14	33.9
1998	252	25.3	62	26.8	10	29.3	349	26.3	14	35.1
1997	160	25.1	61	27.0	15	28.9	333	26.2	16	35.9
1996	144	25.1	86	25.7	7	28.9	228	27.6	11	33.8
1995	54	26.8	124	26.0	4	28.0	165	29.1	13	34.0
1994	101	25.2	128	24.2	11	25.7	208	29.3	8	41.3
1993	80	24.9	132	23.9	13	26.4	188	28.5	10	30.8
1992	66	23.0	93	25.0	6	25.7	163	25.6	9	31.1
1991	33	22.6	44	27.3	2	30.5	147	25.5	8	30.1
1990	26	22.4	35	26.8	2	13.0	104	25.6	9	38.1
Totals & σ	**1,806**	**25.9 5.4**	**2,087**	**27.4 6.3**	**172**	**28.7 3.9**	**3,776**	**27.8 6.6**	**340**	**35.7 9.2**

Fig. 1: Average recycling age (all ships > 499 GT)

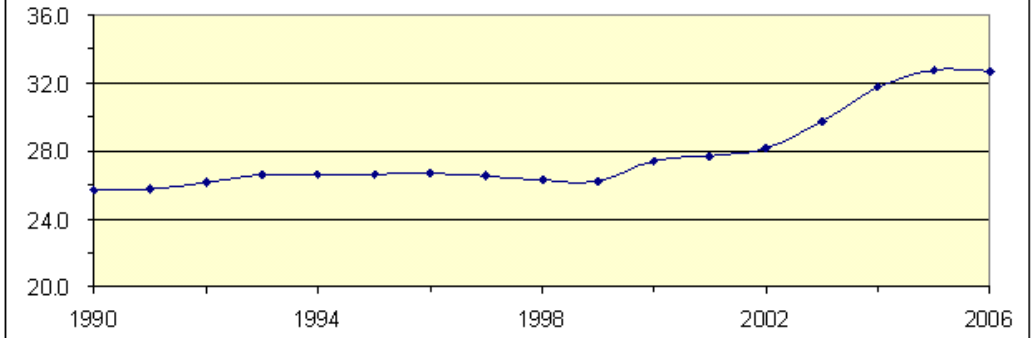


Fig. 2: Tankers' weighted average earnings per day

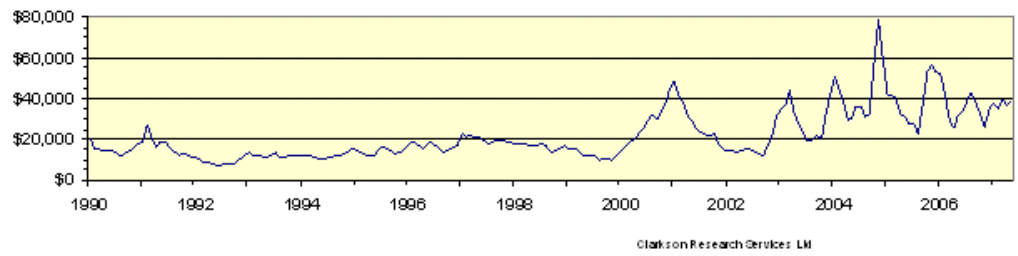


Fig. 3: Baltic Freight Index (Jan 1985=1,000)

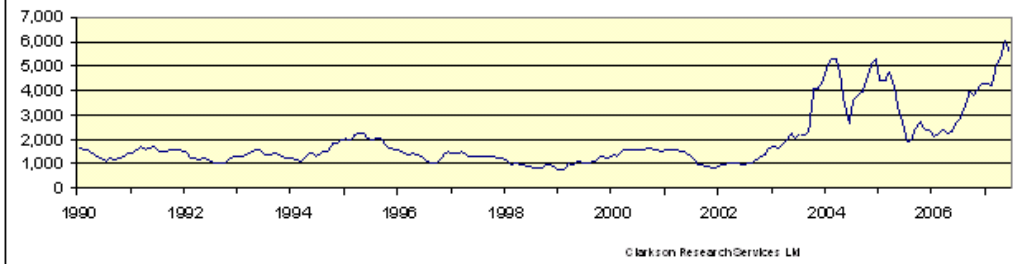
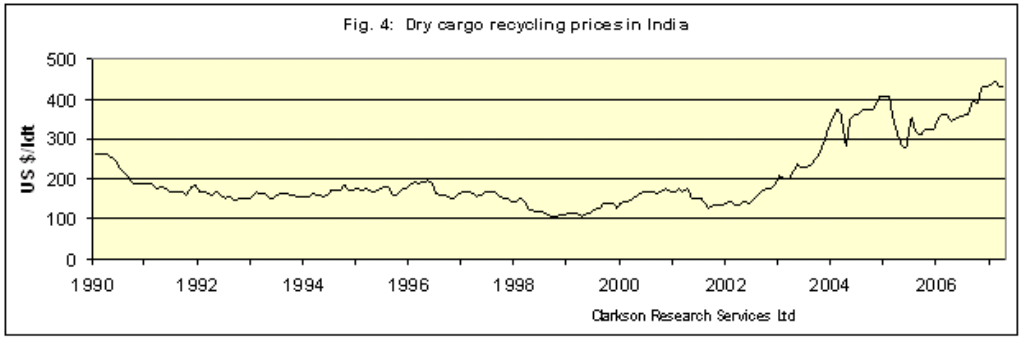


Fig. 4: Dry cargo recycling prices in India



RECYCLING STATE	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	Max 1998-2007	Max 1999-2008
ARGENTINA	-	-	-	461	-	-	-	22,403	-	-	-	22,403	22,403
AUSTRALIA	-	-	-	-	-	466	-	-	4,807	-	103	4,807	4,807
AZERBAIJAN	-	3,752	-	-	-	-	1,467	-	208	-	-	3,752	3,752
BANGLADESH	2,626,359	3,798,540	2,406,923	4,992,402	4,893,973	2,889,908	3,356,537	2,113,765	2,882,503	1,837,988	4,176,026	4,992,402	4,992,402
BELGIUM	-	8,486	-	-	283	-	10,220	1,374	2,064	878	11,554	10,220	11,554
BERMUDA	-	-	-	-	-	-	-	-	-	-	134	-	134
BRAZIL	969	6,405	-	8,680	15,372	6,985	16,490	-	-	-	447	16,490	16,490
BULGARIA	-	-	-	-	1,972	1,827	-	-	-	-	-	1,972	1,972
CANADA	-	21,184	128	-	54,629	17,895	561	-	21,328	8,233	1,495	54,629	54,629
CHILE	-	-	1,314	-	-	8,276	917	-	-	1,110	-	8,276	8,276
CHINA (PEOPLE'S REP. O	550,201	2,564,480	2,637,463	2,509,792	3,138,838	5,582,476	1,538,067	151,089	254,146	340,738	927,762	5,582,476	5,582,476
COLOMBIA	-	-	-	7,522	-	-	-	-	-	6,788	-	7,522	7,522
COOK ISLANDS	-	-	-	-	-	-	-	-	-	912	465	912	912
CROATIA	366	-	-	134	659	1,898	831	889	403	-	242	1,898	1,898
CUBA	-	-	-	13,342	1,843	-	-	-	-	-	-	13,342	13,342
DENMARK	480	11,615	497	309	9,941	15,120	10,674	19,704	15,688	11,538	12,011	19,704	19,704
ECUADOR	-	-	-	-	705	-	131	3,927	-	-	-	3,927	3,927
EGYPT	24,444	8,180	13,695	-	-	8,860	18,299	-	-	-	-	24,444	18,299
ESTONIA	1,182	-	1,266	1,193	-	-	1,163	-	-	239	2,342	1,266	2,342
FALKLAND ISLANDS	-	-	-	-	-	-	-	507	-	-	-	507	507
FRANCE	-	-	-	-	-	-	-	-	-	-	373	-	373
FRENCH POLYNESIA	-	-	-	-	-	-	-	888	-	-	-	888	888
GERMANY	-	-	-	-	-	-	-	1,147	-	-	-	1,147	1,147
GHANA	-	-	-	-	6,629	-	-	-	1,005	898	-	6,629	6,629
GREECE	655	-	-	-	-	7,648	457	-	2,307	10,815	-	10,815	10,815
GUYANA	-	-	-	-	-	-	-	-	-	-	2,230	-	2,230
HONDURAS	-	-	-	-	-	2,556	-	-	-	-	-	2,556	2,556
ICELAND	-	-	-	-	-	-	-	-	144	721	-	721	721
INDIA	6,256,158	6,802,658	5,987,392	4,767,933	6,751,349	5,886,259	1,619,505	1,123,487	852,990	1,332,492	2,458,113	6,802,658	6,802,658
INDONESIA	26,969	1,317	17,577	-	-	-	1,285	209	-	-	2,813	26,969	17,577
IRAN	-	13,865	-	-	-	-	-	-	-	-	-	13,865	13,865
IRISH REPUBLIC	-	-	418	-	-	-	-	-	836	136	-	836	836
ISRAEL	-	-	-	-	-	2,876	-	-	-	-	-	2,876	2,876
ITALY	-	-	2,100	7,354	2,420	6,774	581	-	125	597	-	7,354	7,354
JAPAN	19,212	31,292	22,031	11,521	25,003	45,710	5,767	-	712	-	495	45,710	45,710
KAZAKHSTAN	-	-	-	340	-	-	-	-	-	-	-	340	340
KENYA	-	-	-	1,015	-	988	-	-	-	-	-	1,015	1,015
KOREA (DPR)	-	-	4,234	-	-	-	-	23,281	-	-	-	23,281	23,281
KOREA (REP. OF)	-	7,316	612	-	-	2,303	5,800	467	-	-	-	7,316	7,316
KUWAIT	-	-	-	-	-	153	-	-	-	-	-	153	153
LATVIA	1,451	2,431	629	-	1,386	5,450	1,400	6,976	2,434	11,384	118	11,384	11,384
LEBANON	398	-	-	-	-	999	1,213	-	-	-	-	1,213	1,213
LITHUANIA	-	-	270	2,790	193	1,919	4,297	118	4,584	685	2,699	4,584	4,584
MALAYSIA	332	-	-	-	-	-	-	-	-	-	-	332	-
MALTA	-	-	-	-	-	-	-	-	745	-	-	745	745
MEXICO	40,161	65,775	12,124	1,886	-	1,652	12,753	-	1,396	-	8,671	65,775	65,775
MONTENEGRO	-	-	-	-	-	-	-	-	-	120	-	120	120
MOROCCO	235	-	-	-	-	-	-	-	-	-	-	235	-
MYANMAR	-	-	-	-	-	12,418	-	-	-	-	-	12,418	12,418
NETHERLANDS	1,563	-	463	1,020	4,044	1,919	28,879	4,611	3,403	1,485	6,602	28,879	28,879
NETHERLANDS ANTILLES	-	-	-	-	-	2,927	-	-	1,422	-	-	2,927	2,927
NEW ZEALAND	-	-	-	-	-	-	366	-	-	1,390	3,007	1,390	3,007
NORWAY	646	198	-	244	799	2,614	5,278	1,206	-	489	6,261	5,278	6,261
OMAN	-	-	-	-	-	-	-	-	-	-	574	-	574
PAKISTAN	2,210,276	2,380,488	789,216	1,738,640	997,236	816,961	209,055	47,530	186,987	379,601	273,937	2,380,488	2,380,488
PANAMA	499	-	-	-	-	2,371	-	-	-	-	-	2,371	2,371
PERU	22,087	-	-	7,669	-	18,168	-	-	-	-	-	22,087	18,168
PHILIPPINES	146,871	-	-	-	-	-	3,138	-	-	-	-	146,871	3,138
POLAND	-	-	-	-	-	-	-	1,028	-	-	-	1,028	1,028
PORTUGAL	-	8,721	-	499	-	10,836	-	-	-	1,409	249	10,836	10,836
REUNION	-	-	-	-	964	-	-	-	-	-	-	964	964
ROMANIA	-	-	-	-	-	-	2,613	-	-	-	109	2,613	2,613
RUSSIAN FEDERATION	25,269	10,749	5,446	4,001	6,791	16,805	7,377	3,335	19,404	9,848	12,348	25,269	19,404
SINGAPORE	-	1,406	314	-	-	-	-	697	-	-	-	1,406	1,406
SOUTH AFRICA	444	728	-	-	-	801	-	-	-	802	-	802	802
SPAIN (incl. Canary Isles)	33,459	48,193	26,421	8,605	27,285	32,614	12,633	6,121	6,871	1,041	6,165	48,193	48,193
SWEDEN	-	-	-	504	-	-	1,065	-	-	-	-	1,065	1,065
SYRIA	-	-	-	5,229	-	-	-	-	-	-	-	5,229	5,229
TANZANIA	-	-	-	-	-	-	-	-	-	-	520	-	520
THAILAND	-	3,383	1,217	-	-	-	-	-	-	-	677	3,383	3,383
TURKEY	182,005	225,772	294,541	164,728	385,437	280,367	200,183	137,693	148,448	117,817	141,351	385,437	385,437
TURKMENISTAN	950	-	-	-	-	-	-	-	-	-	-	950	-
UKRAINE	5,620	2,450	1,297	-	1,472	-	-	-	10,165	-	-	10,165	10,165
UNITED ARAB EMIRATES	-	-	-	-	-	-	-	-	-	-	950	-	950
UNITED KINGDOM	1,484	2,710	1,252	1,235	512	2,791	1,491	2,340	2,771	2,535	-	2,791	2,791
UNITED STATES OF AMEI	1,684	-	20,055	140	6,624	26,219	79,139	24,629	79,362	47,490	196,410	79,362	196,410
URUGUAY	797	797	11,188	-	-	-	-	-	-	-	-	11,188	11,188
VIETNAM	73,854	128,886	55,285	33,585	3,912	15,398	8,076	1,898	5,813	3,834	-	128,886	128,886
OTHER	5,789	9,901	14,152	-	-	-	-	-	-	-	-	14,152	14,152
UNKNOWN	21,804	10,801	2,486	2,708	36,390	185,985	27,784	70,891	78,574	18,834	22,534	185,985	185,985
TOTALS	**12,284,673**	**16,182,479**	**12,332,006**	**14,295,481**	**16,376,661**	**15,928,192**	**7,195,492**	**3,772,210**	**4,591,645**	**4,152,847**	**8,279,787**	**21,332,879**	**21,289,147**

2008	World fleet	830,704,412
	40% of World	332,281,765
	3% of 40%	**9,968,453**